BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Logic Gates: AND, OR and NOT

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What must be true before binary search is used?

2. What does software copyright restrict?

A) Using any keyboard
B) Learning to code
C) Saving your own work
D) Copying without permission

3. "Proprietary software always gives every user unrestricted source-code modification rights." Correct this. Explain why.

4. A ticket program follows the language's rules but adds 5 instead of multiplying by 5. "Logic errors always stop translation." Correct this. How should the error be found?

Today's objectives

Recognise gate symbols and explain each gate. Predict outputs for specified input values.

Key vocabulary - Read, cover, recall

AND
Output 1 only when both inputs are 1.

OR
Output 1 when either or both inputs are 1.

NOT
Reverses one Boolean input.

Gate
Electronic circuit implementing a Boolean rule.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Binary logic	Logic gates work with Boolean input and output values 0 and 1.
AND gate	AND outputs 1 only if both inputs are 1. Its symbol has a flat input edge and rounded output edge.
OR gate	OR outputs 1 if either or both inputs are 1; it outputs 0 only for 0,0.
NOT gate	NOT has one input and outputs the opposite value; its symbol has a small inversion bubble.

I Do - One worked example

An alarm sounds only if a door is open (A=1) AND the system is armed (B=1). Explain the outputs for A=1,B=1 and A=1,B=0. [4 marks]

1. This is an AND rule because both conditions must be true [1].
2. With 1,1 the output is 1 [1], so the alarm sounds.
3. With 1,0 the output is 0 [1] because both are not true [1].
4. Show each input pair alongside its output before revealing.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A buzzer sounds if either A or B is on. Give OR outputs for A,B = 00,01,10,11. [4 marks]

2. A safety indicator is NOT A. Give output for A=0 and A=1. [2 marks]

3. A fan needs power A and permission B. Explain why AND is suitable and state outputs for 1,0 and 1,1. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the Boolean test so both security checks must be true.

```
signedIn = true
hasPermission = true
if ... then
print("Allow")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the output of an AND gate given A=1 and B=0. [1 mark]

Q3. Give OR outputs for input pairs 00, 01, 10 and 11 in that order. [4 marks]

Q4. Explain AND, OR and NOT with TWO different inputs or examples for each. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An OR gate outputs 1 when both its inputs are 1. ☐ T ☐ F
2. Logic gates work with Boolean input and output values 0 and 1. ☐ T ☐ F
3. AND outputs 1 only if both inputs are 1 ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The list must be sorted. This lets the search safely rule out half the remaining items. ☐

Q2 D - Copying without permission ☐

Q3 Source code is normally controlled by its owner and users receive specified usage rights. ☐

Q4 A logic error gives the wrong result even when the code follows the language's rules. Fix syntax errors first, then trace and test the logic. ☐

Code First - Model

```
signedIn = true
hasPermission = true
if signedIn AND hasPermission then
print("Allow")
endif
```

Output: Allow

We Do - Check the model

1. 0,1,1,1 in the given order, one mark for each correct output. ☐

2. For A=0 output 1 [1]; for A=1 output 0 [1]. ☐

3. Both inputs must be high [1], so AND is suitable [1]. 1 AND 0=0 [1]; 1 AND 1=1 [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 0 [1]. ☐

Q3 0 [1], 1 [1], 1 [1], 1 [1]. ☐

Q4 AND 1,1 -> 1 [1] and 1,0 -> 0 [1]. OR 0,1 -> 1 [1] and 0,0 -> 0 [1]. NOT input 0 -> 1 [1] and input 1 -> 0 [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: